

# **life-line Documentation**

***Versión 1.0.0***

**© 2023, Michael Altfield and the Eco-Libre Team**

junio 03, 2026



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Welcome! This is the official documentation for the Eco-Libre Life-Line.

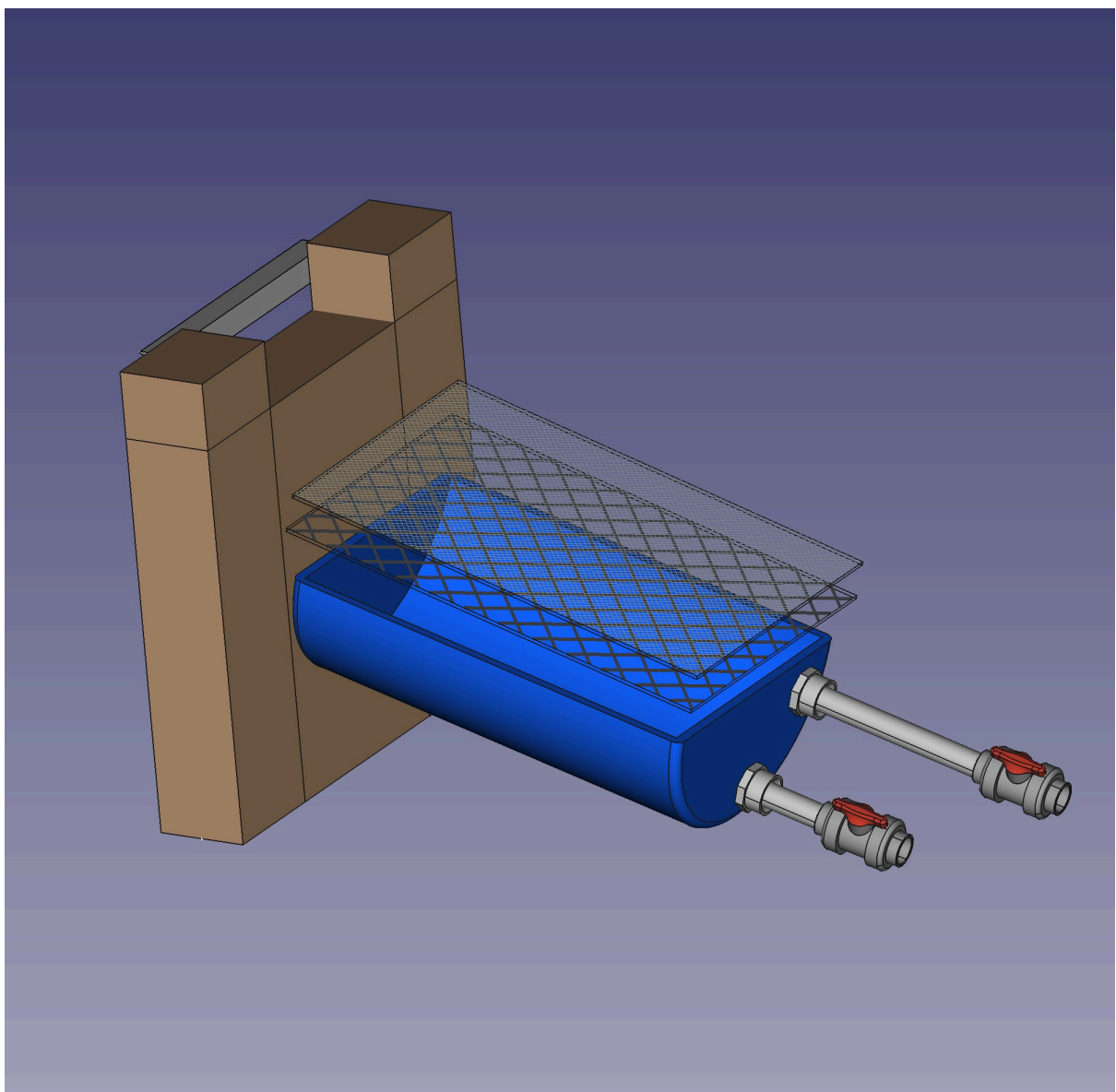


Figura 1. CAD Design of the Eco-Libre Life-Line

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**Nota** Just want to know **how to build the Eco-Libre Life-Line?**

See our [Capítulo 5](#)

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**Introduction**

---

The Eco-Libre Life-Line is a raw water intake, filtration, and storage system.



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## Theory

---

This section will describe the research behind this project and how it would benefit the lives of your community.

### 2.1 Concept

The Eco-Libre Life-Line project is a system designed to collect, filter, and store surface water for an off-grid community.

### 2.2 Design Goals

The water system should be:

1. Low-Maintenance
2. Hygienic

### 2.3 Low-Maintenance

**Low-Maintenance** means that low-tech automation and well-engineered, self-cleaning systems should be employed so that the system only needs routine maintenance once or twice per year.

### 2.4 Hygienic

**Hygienic** means that the water stored in the clean water tanks and delivered to sinks should be free of contamination (bacteria, parasites) and be safe to drink.

### 2.5 Size

**Pico (3-person)** (the smallest Eco-Libre Life-Line water system) should provide sufficient water for 3-people.

**Nano (15-person)** The small-size system should be horizontally scalable by multiplying the number of filters and tanks to provide sufficient water for a 15-person community

**Micro (30-person)** To handle a community of up to 30-people, a larger system design is provided.

**Milli (150-person)** To handle a community of up to 150-people, the micro system is horizontally

scaled.

## 2.6 Components

### 2.6.1 Intake

The water intake will consist of a weir on a stream with two openings with turbulence bars that channel turbulent water down onto a screen-covered 220-liter barrel angled at a downward slope.

Error opening image file: cannot identify image file '/\_\_w/life-line/life-line/docs/images/life-line\_intake\_parts.svg'

Figura 2.1. CAD Design of the Eco-Libre Life-Line's «Intake» component

This is a common low-budget, self-cleaning intake design that will shed organic debris during high-flow (eg a heavy rain).

The distance from the bottom of the weir to the top of the screen should be at least 20 cm.

The size of the holes in the screen should be 1-5 mm (it should be supported by an expanded metal screen below it).

The screen on the barrel should be mounted on a hinge to allow for routine maintenance.

### 2.6.2 Settling Tank

A radial-flow settling tank should be located very near to the intake and will remove sediment that passed the intake screen.

### 2.6.3 Large Sediment Filter

A self-cleaning spin down filter (largest size micron available) will remove debris that passed through the settling tank.

The flush valve should be automatically opened electronically by a timer.

### 2.6.4 Pre-Filter

This pre-filter will remove turbidity from the water before it enters the slow sand filter.

This is built from a 220-litre HDPE drum that's filled with rocks and (mostly coarse) sand.

### 2.6.5 Slow Sand Filter

The Slow Sand Filter will remove harmful bacteria and parasites from the water.

This is built from a 220-litre HDPE drum that's filled with rocks and (mostly fine) sand.

### 2.6.6 Media Sediment Filter

A self-cleaning spin down filter (smallest size micron available) will remove filtering media debris (sand) that may have come-out of the filters

The flush valve should be automatically opened electronically by a timer.

### 2.6.7 Clean Water Tank

The clean water tank stores potable water.

### 2.6.8 Lifeline

Pipes supply clean water from the clean water tanks to the community.



This section will show example builds of the Eco-Libre Life-Line around the world.

### 3.1 Boris Plotkin

In Nov 2024, [Boris Plotkin](#) built the first prototype of the Life-Line at the [Fruit Haven](#) community in Morona-Santiago, Ecuador



Figura 3.1. Concrete Block, Rock, and Cement Weir for the Eco-Libre Life-Line at Fruit Haven in Morona-Santiago, Ecuador (attribution: [Boris Plotkin](#))

### 3.2 Vegan Beasts

In Apr 2025, [Vegan Beasts](#) built the second prototype of the Life-Line at the [Fruit Haven](#) community in Morona-Santiago, Ecuador





Figura 3.2. Rock and Cement Weir for the Eco-Libre Life-Line at Fruit Haven in Morona-Santiago, Ecuador (attribution: [Vegan Beasts](#))

### 3.3 Get Listed!

Did you build this? Please [let us know](#) and we'll add your build to this page.



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## Frequently Asked Questions

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### 4.1 Q: How can I get updates about Eco-Libre?

You can [signup](#) for our Eco-Libre email newsletter or follow us on any of the following social media profiles:

- Twitter [@EcoLibreNews](#)
- Facebook [@EcoLibreNews](#)
- Mastodon [@ecolibre@chos.social](#)
- GitHub [Eco-Libre](#)
- YouTube [@Eco-Libre](#)

### 4.2 Q: How much does it cost to build this system?

See [Section 5.1.2](#)

### 4.3 Q: What do I need to buy to build this?

See [Section 5.1.2](#)

### 4.4 Q: Where can I find the instructions to actually build it?

See our [Capítulo 5](#)

### 4.5 Q: Has anyone else built this before?

Yes. See [Capítulo 3](#)

### 4.6 Q: Where can I download the 3D Models?

You can download our source [FreeCAD](#) files from our [GitHub](#) repo.

## 4.7 Q: Can I sell this?

Please do!

All of Eco-Libre's work is libre or open source, pursuant to the [Open Source Hardware Association definition](#).

You're welcome to take our designs and use them for commercial purposes, so long as

1. You provide prominent attribution back to Eco-Libre (link to [our website](#)) and
2. You must make public any modifications that you make to our work, and you must also license your derivative works under a compatible ShareAlike license.

For more information, see [Capítulo 9](#)

## 4.8 Q: My questions isn't listed. Where can I get support?

Please see [Capítulo 7](#)

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## Getting Started

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This section will document how to build the Eco-Libre Life-Line.

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**Nota** You can view the documentation to build a previous version by selecting a different version in the bottom-left menu

---

The Eco-Libre Life-Line project is made up of several distinct components. We have separated the build instructions into different sections to build each component.

Click on the component below to view its build instructions.

### 5.1 Intake

This section will describe how to build the «Intake» component of the Eco-Libre Life-Line system.

#### 5.1.1 What is the «Intake» Component?

This section will describe how to build the «Intake» component of the Eco-Libre Life-Line system.

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Figura 5.1. CAD Design of the Eco-Libre Life-Line's «Intake» component

1. Weir 14
2. Turbulence Bar 14
3. Fine Mesh Screen 15
4. Expanded Metal Mesh 16
5. Outflow Pipe 16
6. Cleanout Drain Pipe 17
7. Drum 18

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**Nota** Want to **view the 3D Model** of the Intake Component above?

You can download the `intake.FCStd` file from our [GitHub repo](#) and open it on your computer in [FreeCAD](#)

---

The «Intake» component is responsible for raw water capture from a stream. The design of  $\geq 20\text{cm}$  falling, turbulent water is self-cleaning: the turbulent water falls onto the screen with sufficient force to clear debris off of the screen. This ensures constant raw water collection into the Life-Line system, without requiring periodic manual human intervention (or automatic moving parts) needed to remove debris blocking the ingest of water into the system.

The «Intake» component is made-up of several parts:

## 1. Weir

The Weir is a masonry structure (CEB, Concrete Block, or Steel-reinforced Concrete) that's built in the middle of a stream. It has a Weir Opening at the top, which channels falling water to flow ontop of a metal screen and into a water collection drum.



The distance between the bottom of the Weir Opening and the top of the Fine Mesh Screen atop the water collection drum must be at least 20 cm. If the water falls less than 20 cm, it will not have enough force to self-clean the screen of organic debris.

## 2. Turbulence Bar

The Turbulence Bar is a piece of steel angle that's bolted to the weir, bisecting the Weir Opening.

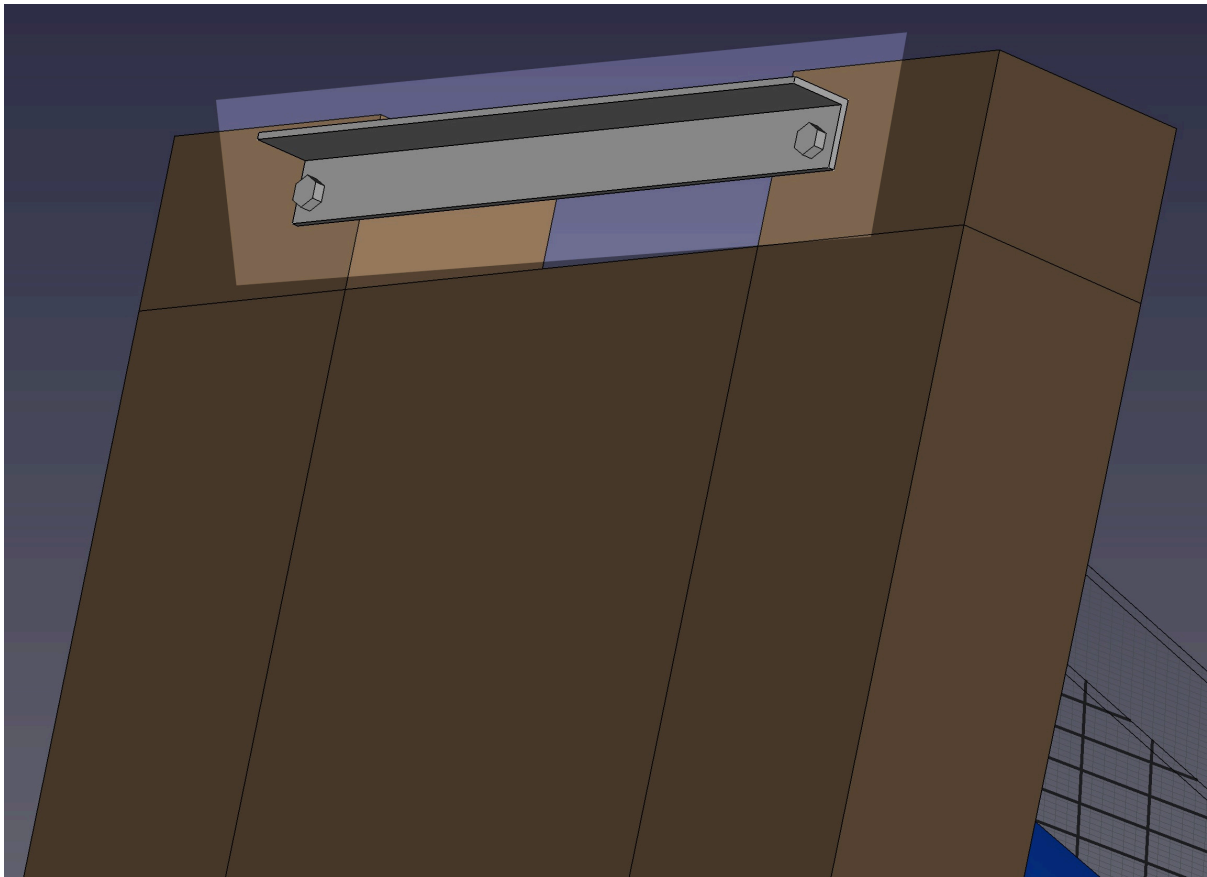


Figura 5.2. The Life-Line «Intake» component's Turbulence Bar

The Turbulence Bar changes the flow of the water as it passes through the Weir Opening, creating turbulence. This turbulent water is more effective at self-cleaning the [screen](#) below of organic debris than non-turbulent falling water.

### 3. Fine Mesh Screen

The Fine Mesh Screen is a very fine metal screen that's designed to filter-out large organic materials (eg leaves, sticks, stones, grass, etc) from entering the [drum](#).



The Fine Mesh Screen should be made of Stainless Steel, and the holes in the screen should be approximately 0.85 mm large (20 [US Mesh](#))

#### 4. Expanded Metal Mesh

The Expanded Metal Mesh is a strong grate made of Expanded Metal that sits below the [Fine Mesh Screen](#) and atop the Drum

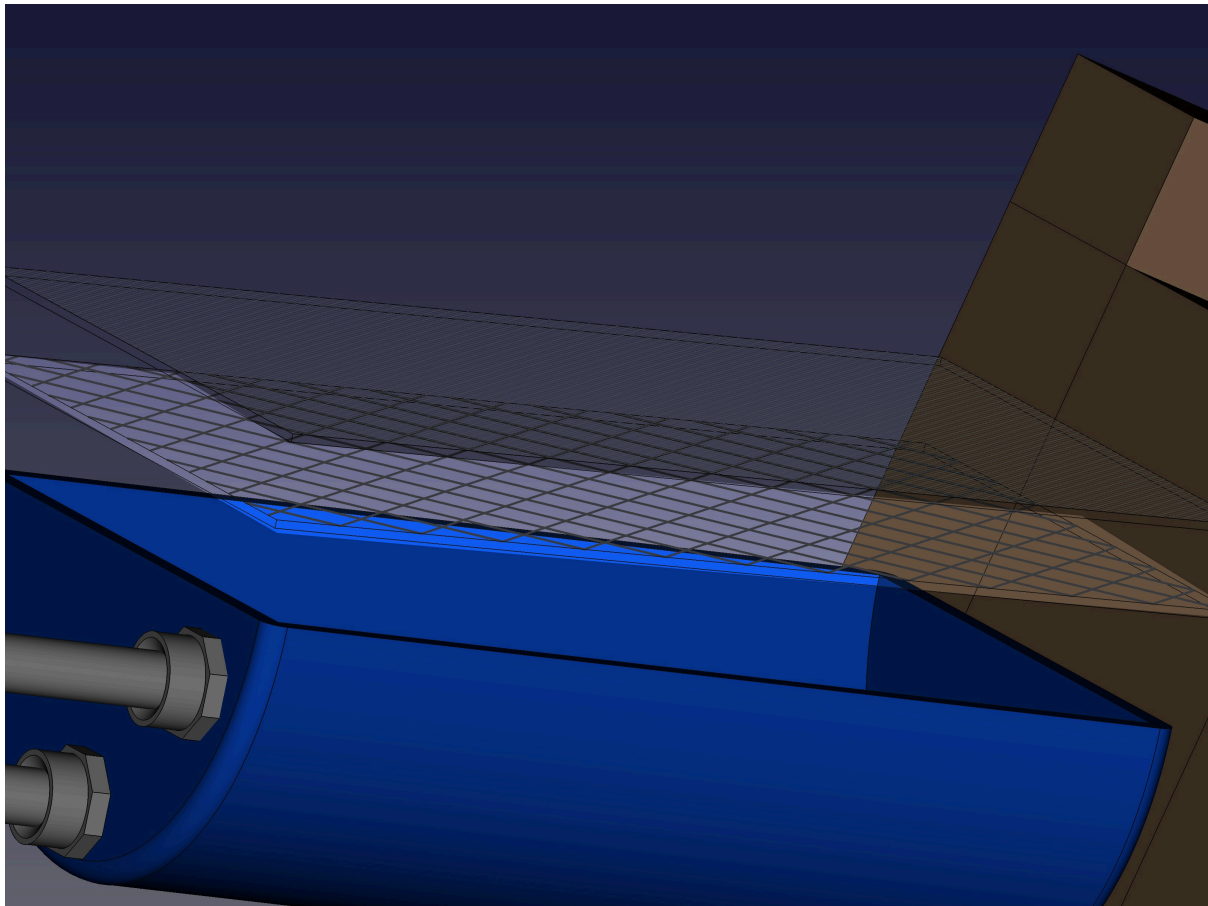


Figura 5.3. The Life-Line «Intake» component's Expanded Metal Mesh

The purpose of the Expanded Metal Mesh is to provide a physical support structure for the [Fine Mesh Screen](#) above it.

#### 5. Outflow Pipe

The Outflow Pipe is where water flows out of the Intake [drum](#) and into the next component of the Eco-Libre Life-Line system (the [Section 5.2 Component](#))

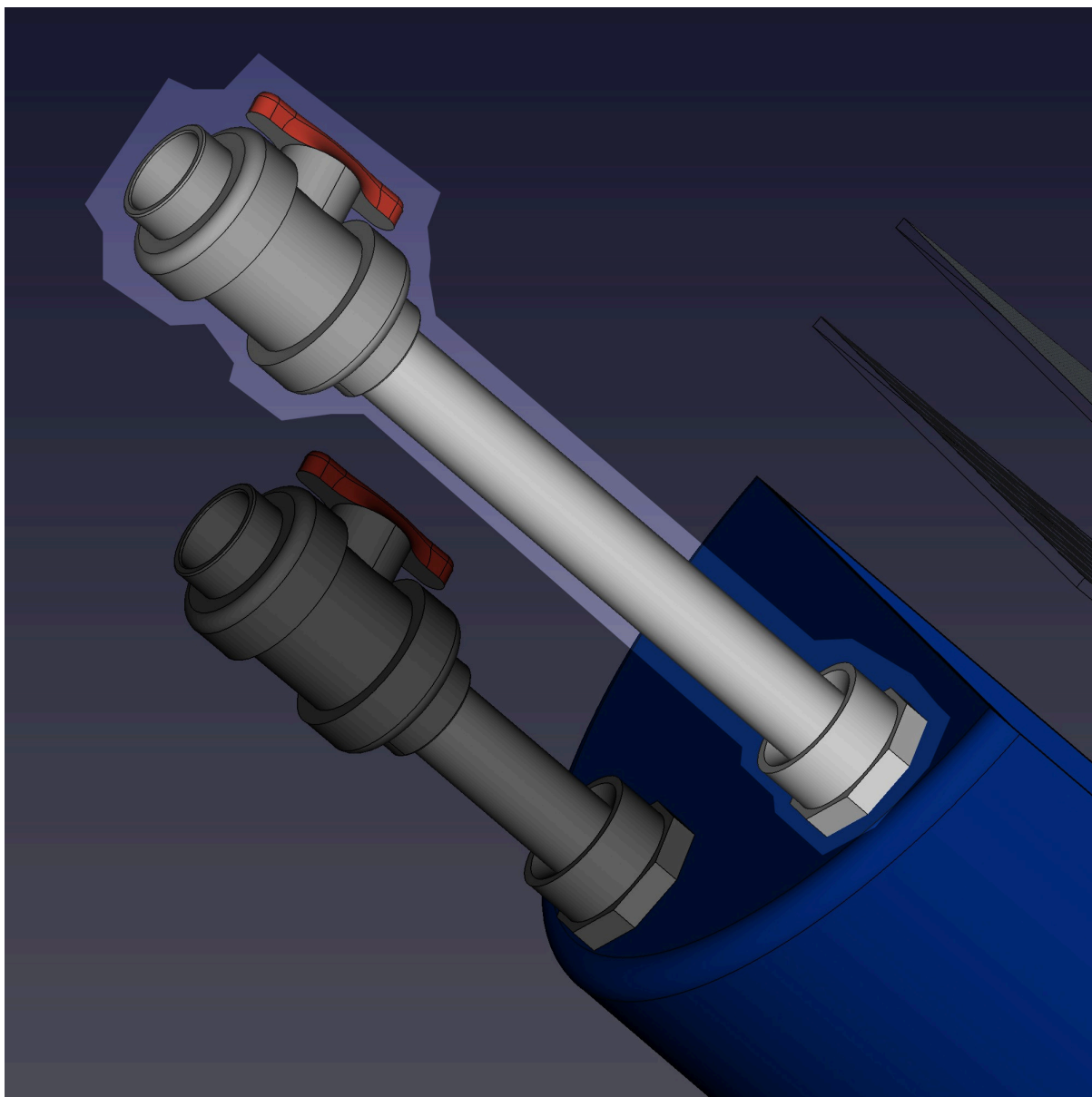


Figura 5.4. The Life-Line «Intake» component's Outflow Pipe

The outflow pipe consists of a Bushing, Pipe, and Valve.

#### **6. Cleanout Drain Pipe**

The Cleanout Drain Pipe is located at the bottom of the **drum**, next to the **Outflow Pipe**.



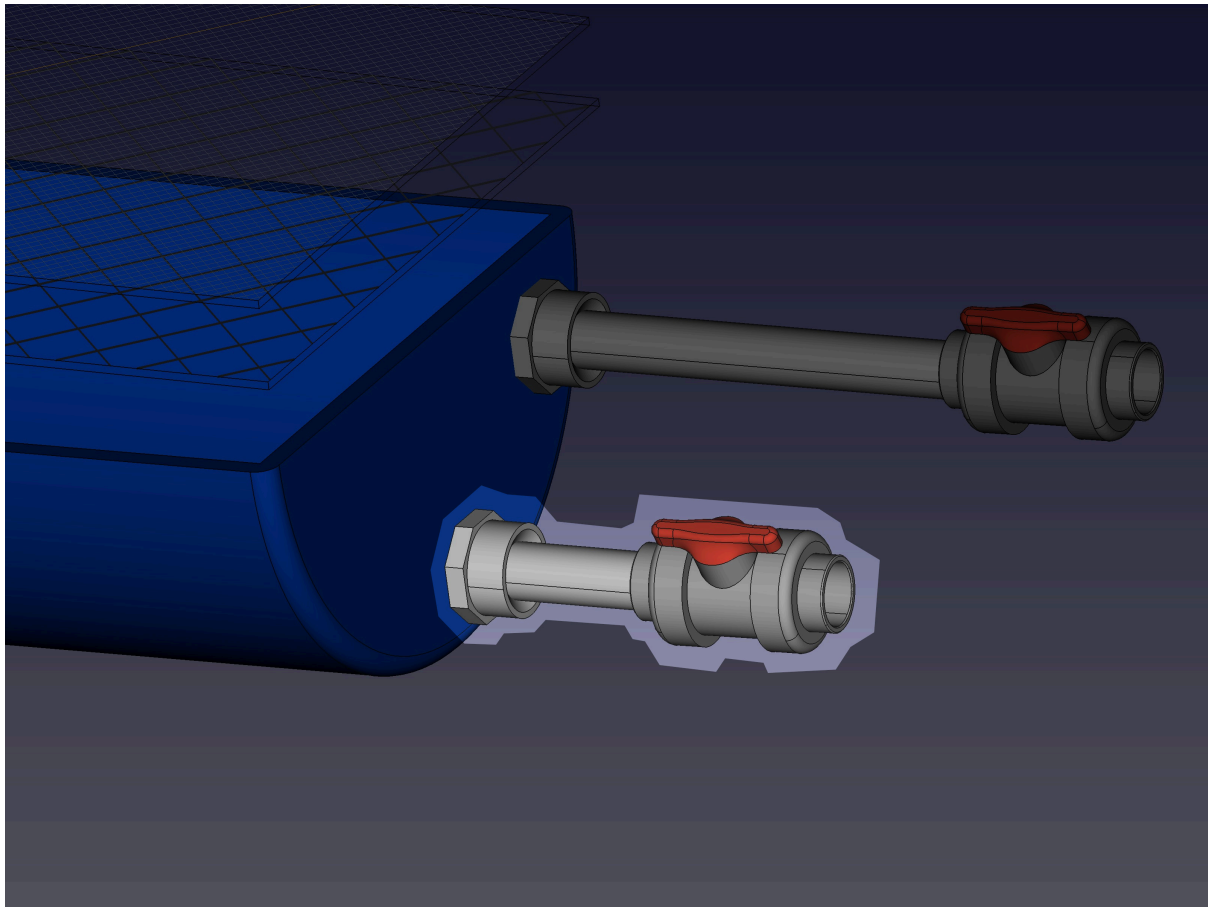


Figura 5.5. The Life-Line «Intake» component's Cleanout Drain Pipe

The Cleanout Drain Pipe consists of a Bushing, Pipe, and Valve.

The purpose of the Cleanout Drain Pipe is to flush small debris (eg sand) that have collected at the bottom of the drum. In normal operations, the valve should be closed. During routine maintenance (at least once per year), the Cleanout Drain Pipe should be opened, to flush-out any debris that have settled at the bottom of the drum.

If possible, a valve actuator (eg a simple electronic sprinkler system on a timer) could be placed after the valve to automatically flush the drum on a routine basis.

## 7. Drum

The Drum is a standard 220L (55 gallon) HDPE barrel drum that's been cut in half length-wise.



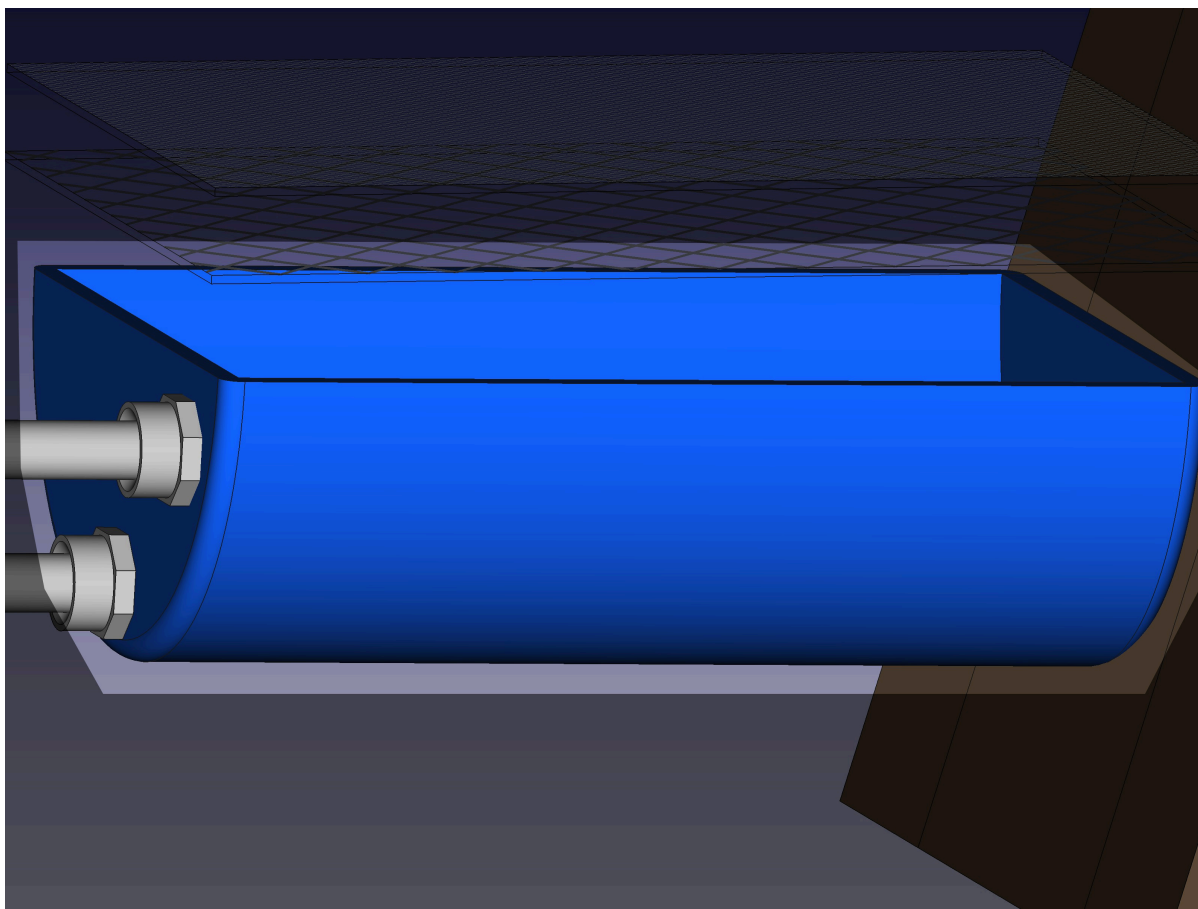


Figura 5.6. The Life-Line «Intake» component's Drum

The drum collects water as it spills over the weir and onto the [metal mesh screen](#) that sit atop the cut-open face of the drum.

Two holes are cut on the downstream side of the drum, where through-wall bushings are placed for the [Cleanout Drain Pipe](#) and the [Outflow Pipe](#).

### 5.1.2 Bill of Materials

This section will list the materials needed to build the Intake component of the Eco-Libre Life-Line project.

| Qty | Item | Price (EUR) | Material                                                                | Weight (kg) | Dimension (mm)    | Capacity | Tolerances | Product URI (with price)                                                                                                                                                      | Other Notes                                        |
|-----|------|-------------|-------------------------------------------------------------------------|-------------|-------------------|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 2   | Weir | 759.69      | Rocks, CEBs, Concrete Blocks, Bricks, and /or Steel-Reinforced Concrete |             | 1035 x 1000 x 500 |          |            | <a href="https://brooksonline.ie/kil-saran-kpro-crete-multi-purpose-concrete-25kg-un0427">https://brooksonline.ie/kil-saran-kpro-crete-multi-purpose-concrete-25kg-un0427</a> | Highly variable; price is minimum in just concrete |

|                                          |                    |                  |     |      |                              |                                                                                                                                                                     |                                                                                                                                                                                                                                             |  |
|------------------------------------------|--------------------|------------------|-----|------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1 Drum                                   | 109                | HDPE             | 5.5 | Blue | 493<br>745                   | 220<br>Liters                                                                                                                                                       | <a href="https://www.denios.ie/plastic-drums-with-bung-hole-120-220-litre-volume-10300300451400/266140">https://www.denios.ie/plastic-drums-with-bung-hole-120-220-litre-volume-10300300451400/266140</a>                                   |  |
| 2 Steel Angle                            | 11.825666666666667 | Galvanized Steel |     |      | 650<br>x<br>60<br>x<br>6     |                                                                                                                                                                     | <a href="https://mybuildingsupplies.ie/shop/building-supplies/steel/steel-beams-steel-lengths-steel-sheets/steel-angle/">https://mybuildingsupplies.ie/shop/building-supplies/steel/steel-beams-steel-lengths-steel-sheets/steel-angle/</a> |  |
| 4 M16 Concrete Anchor Bolt               | 8.664              | Galvanized Steel |     |      | M16<br>x<br>100<br>mm        |                                                                                                                                                                     | <a href="https://www.screwfix.ie/product/rawlplug-through-bolts-r-xpt-m16-x-100mm-25-pack/657vc">https://www.screwfix.ie/product/rawlplug-through-bolts-r-xpt-m16-x-100mm-25-pack/657vc</a>                                                 |  |
| 2 Fine Mesh Screen                       | 17.1792010250685   | Stainless Steel  |     |      | 1000<br>x<br>665             | 0.85 MVOYTLARG-304-stainless-steel-mesh-47in-x-10ft-20-mesh-nev-large-rust-sturdy-metal-mesh-screen-work-even-wire-mesh-roll-for-window-screen-replacement-door-air |                                                                                                                                                                                                                                             |  |
| 2 Expanded Metal                         | 186.067            | Galvanized Steel |     |      | 1000<br>x<br>665             |                                                                                                                                                                     | <a href="https://www.lenehans.ie/steel-expanded-metal-sheet-1000mm-x-500mm.html">https://www.lenehans.ie/steel-expanded-metal-sheet-1000mm-x-500mm.html</a>                                                                                 |  |
| 4 Through-Wall Bushing (aka "bulk-head") | 120.68             | PVC              |     |      | 50<br>mm<br>(2<br>inch)      |                                                                                                                                                                     | <a href="https://www.recifart.com/en/fitting/572-bulkhead-50mm.html">https://www.recifart.com/en/fitting/572-bulkhead-50mm.html</a>                                                                                                         |  |
| 2 50 mm Pipe                             | 15                 | PVC              |     |      | 1000<br>mm<br>+<br>500<br>mm |                                                                                                                                                                     | <a href="https://percolationsupplies.ie/product/50mm-pvc-piping/">https://percolationsupplies.ie/product/50mm-pvc-piping/</a>                                                                                                               |  |
| 4 50 mm Valve                            | 55.32              | PVC              |     |      |                              |                                                                                                                                                                     | <a href="https://www.peservices.ie/product/ball-valve-pvc-u-50mm">https://www.peservices.ie/product/ball-valve-pvc-u-50mm</a>                                                                                                               |  |

**Nota** Want to **view the original spreadsheet** of the BOM above?

You can download the docs/build/intake/bom.ods file from our [GitHub repo](#) and open it on your computer in [LibreOffice](#)

## 5.2 Settling Tank

This is a section that will talk about how to build the «Settling Tank» component of the Eco-Libre Life-Line system.

### 5.2.1 What is the «Settling Tank» Component?

This is a section that will talk about how to build the «Settling Tank» component of the Eco-Libre Life-Line system.

Error opening image file: [Errno 2] No such file or directory: '/\_\_w/life-line/life-line/docs/images/life-line\_settling-tank\_parts.svg'

Figura 5.7. CAD Design of the Eco-Libre Life-Line's «Settling Tank» component

---

**Nota** Want to **view the 3D Model** of the Settling Tank Component above?

You can download the `settling-tank.FCStd` file from out [GitHub repo](#) and open it on your computer in [FreeCAD](#)

---

The «Settling Tank» component is responsible for raw water capture from a stream. It's made-up of several parts:

**1. TODO**

TODO

**2. TODO**

TODO

## 5.2.2 Bill of Materials

This section will list the materials needed to build the Settling Tank component of the Eco-Libre Life-Line project.

TODO

For an overview of what each component is, see [Section 2.6](#)



---

# Contributing

---

Thank you for considering to help us!

As an open-source project, we depend on the work of volunteers to stay alive, and your time helping to improve our project is very much appreciated :)

## 6.1 Wish List

Below is a list of items on our TODO list. If you'd like to help us tackle these tasks, please [contact us](#) as there may be someone already working on a given task that you may be able to collaborate with.

Please be aware that any contributions you make will be made open-source. See [Capítulo 9](#) for more info.

TODO

## 6.2 Donate

Running the Eco-Libre project has many fees, including

1. Server & domain infrastructure fees
2. Purchasing hardware items for testing & iterating

If you have more money than free time, you can also help Eco-Libre by making a donation. We accept Bitcoin and Monero.

|                                        |
|----------------------------------------|
| BTC 1DXyJpmu2KQMw2v4QJVzzjZo6f87BBndu6 |
|----------------------------------------|

:: TODO: Open-collective

Donations of any amount are **greatly** appreciated. Thank you!



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## Support

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As an open-source project, we try to keep all of our support requests on GitHub. If you

1. have a question that is not answered by this documentation,
2. discover a bug in the software, or
3. would like to request a feature in the software

then please [create an issue](#) on our GitHub.

If you don't want to register for an account on GitHub or you'd like to remain anonymous, then you can [email us](#). We'll put the issue on GitHub for you.





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## Changelog

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### Eco-Libre Life-Line Changelog

[v2025.10]

<https://www.eco-libre.org/life-line-2025-10-released/>

- \* first formal release (very alpha)
- \* changed expanded metal to hatch of solid (optimization)  
<https://www.eco-libre.org/expanded-metal-hatch-freecad/>
- \* simplified other objects, reducing (compressed) intake.FCStd **from** 15 MB to 0.212 KB
- \* reduced total (uncompressed) MemSize about 172x **from** 114.8 MB to ~0.7 MB
- \* added fine metal mesh (**as** hatched solid)
- \* thank you to Michael Altfield, Jack Nugent, Boris Plotkin, **and** Vegan Beasts



## License

All of Eco-Libre Project's content (including this documentation) is dual-licensed. All code is [GPLv3](#) and all other content is [CC BY-SA](#).

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Version 3, 29 June 2007

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For the developers' and authors' protection, the GPL clearly explains that there is no warranty for this free software. For both users' and authors' sake, the GPL requires that modified versions be marked as changed, so that their problems will not be attributed erroneously to authors of previous versions.

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Finally, every program is threatened constantly by software patents. States should not allow patents to restrict development and use of software on general-purpose computers, but in those that do, we wish to avoid the special danger that patents applied to a free program could make it effectively proprietary. To prevent this, the GPL assures that patents cannot be used to render the program non-free.

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```
<one line to give the program's name and a brief idea of what it does.>
Copyright (C) <year> <name of author>
```

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the Free Software Foundation, either version 3 of the License, or
(at your option) any later version.
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```
<program> Copyright (C) <year> <name of author>
This program comes with ABSOLUTELY NO WARRANTY; for details type `show w'.
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```

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## Attribution

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**Thank you to the following folks who have contributed to the Life-Line project!**

- [Michael Altfield](#)
- [Jack Nugent](#)
- [Boris Plotkin](#)
- [Vegan Beasts](#)

**This project would not have been possible without the following tools and services:**

- [Python](#)
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### 10.1 Similar Projects

The reader may be interested to investigate the following similar projects:

- [Gaviotas](#)







